The problem of online hate speech: extent, automatic detection and conflict over freedom of speech

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Preface

The Bachelor of Arts and Sciences at University College London allowed me to deepen my interest in many disciplines and develop an interest in new ones. For this final project, I wanted to do something challenging, useful and consistent with my academic background. This wish led me to the project to explore the mechanisms of online hate, its detection by algorithms, and finally to create a machine learning model detecting online hate. I was then able to use this model as an analytical tool in several use cases.

During my studies, I developed a taste for computer sciences, to the point of making it the main discipline of my Bachelor's degree. I followed theoretical courses (on machine learning for example), and applied courses (like learning how to code in Python language). Therefore, it was a personal ambition to apply my theoretical knowledge of machine learning to the skills I had acquired. The logical continuation of progress within the discipline was hence to create a machine learning model.

I am motivated to use the powerful tools given by computer sciences to carry out useful and relevant projects. My group project for the BASC0005 module (Quantitative Methods 2: Data Science and Visualisation) is a good example of such an application of computer sciences. In this project, I used public data to demonstrate the causal link between the economic situation of a household and its access to a good education in the Paris region. I coded effective visuals for this, including an interactive map. This project demonstrated the economic inequalities within public education, leading to social determinism, although France has a policy of providing equal education for all.

Beyond my taste for computer science, I also have a strong taste for humanities, as evidenced by the various Philosophy modules I have undertaken these last years. Recently, putting freedom of expression back into perspective of the growing online hate is a fascinating question to take under a philosophical approach.

Moreover, the subject of online hate is linked to a sector of computer science that I have always wanted to explore: natural language processing (NLP). As a lover of literature and language, this field of making a machine understand language is particularly exciting, and succeeding in creating an NLP project is a challenge I wanted to give myself for this essay.

Thus, I concluded that exploring online hate, its detection, and creating a model to apply to relevant use cases seemed to be a particularly exciting topic for my dissertation. Moreover, this essay topic fulfils the criterion of interdisciplinarity. I am going to go beyond computer science and look at the philosophical, social and psychological aspects of online hate, in order to provide a holistic and relevant approach to this vast subject. Finally, if the exploration of ever more efficient methods to detect hate is a widely explored topic nowadays, it is rarely used for analytical purposes, so there is a real gap to study.

Abstract

Online hate speech is a growing and contagious problem with serious social, psychological and violent consequences. In western democracies, there are two opposing philosophies of freedom of speech on this issue: the American libertarian philosophy and the European restrictive philosophy. This disagreement is reflected in the legislation between the two geographical areas, as well as in the tensions between American societies and the European Union.

To further explore the topic of combating online hate speech, this dissertation focuses on its automatic detection by algorithms. In this paper, I have created an online hate detection tool that can detect toxic hate, obscene hate, and offensive hate. This tool is based on a dataset of labelled Wikipedia comments and uses NLP and supervised machine learning techniques. This tool has a balanced accuracy score of around 85% and an f1 score of around 68%. This model was trained and used on the purpose of analyzing online hate, specifically on the question of freedom of speech.

Therefore, I applied the model to four datasets: a dataset of tweets from MPs during the Brexit, a dataset of tweets with the #politics, a dataset of tweets about the coronavirus, and a random dataset serving as a reference. These use cases implied the beginnings of a correlation between an area's legislation and its online hate rate. If the aim is to reduce online hate, this correlation would be an argument in favour of the European vision and legislation of freedom of expression. Nevertheless, the limited scope of our study and certain biases prevents us from asserting the existence of a strong correlation or a causal relationship.

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1 Introduction

The fight against hate speech is a challenging issue which entails complex political, ethical, and technical dilemmas. Nowadays, this problem has a new dimension due to internet where online hate speech proliferates. Hate speech is defined by Encyclopædia Britannica as "speech or expression that denigrates a person on the basis of membership in a social group"[1].

Online hate speech directly affects the lives of victims, whether it is their inclusion in society or their psychological state. However, the fight against online hate is complex because it confronts different views of freedom of speech. These different philosophies are reflected in conflicting legislation within states, making it difficult to implement effective measures. In addition, the uncontrollable nature of cyberspace limits the impact of the solutions implemented.

Reducing online hate speech implies its automatic detection by algorithms that use Natural Language Processing and machine learning techniques. Natural Language Processing consists in making algorithms process human language. Machine learning consists in making an algorithm mimic the way humans learn with the use of data; in this case, it will try to learn to detect online hate.

In this dissertation, I will attempt to create a model that can detect not only online hate speech but also the type of hate. The literature review will first look at the extent, consequences, issues, and legislation of online hate speech. Secondly, the literature review will explain the techniques required to create this model.

Once this detection tool has been created, it will be used in several use cases as an analytical tool to lead to relevant discussions about online hate speech. These use cases will look at the link between online hate speech and different visions of freedom of expression online.

2 Literature Review

2.1 Online hate speech

2.1.1 A growing and contagious problem

Growth of online hate speech

With the emergence of social networks, online hate speech became particularly prevalent. In 2020, 41% of Americans say they have experienced online hate, compared to 35% in 2017 [2]. The number of physical threats online has doubled from 7% to 14% of Americans over the same period. Certain sensitive topics are particularly prone to online hate: gender, sexuality, religion, race/ethnicity and politics. Politics is the

number one topic for online hate in the US, accounting for more than 50% of Americans experiencing online hate in 2020.

The growth of online hate was especially noticeable during the Covid crisis, with an increase in the use of internet as social and cultural life was online. In addition, the crisis has led to many tensions, polarising populations and leading to more online hate. In 2021 for example, online hate increased by 20% during the pandemic in the US and UK[3].

Hate speech is a type of online content that spreads much faster than other types of content. Studies about the spread of hate speech concluded that hateful content spreads faster, farther, and reaches a wider audience than non-hateful content, particularly as users of hateful content are highly interconnected and relayed [4].

Contagion of online hate speech

The ability of online hate speech to spread quickly can be explained by its contagious aspect. Various psychological mechanisms entail that contact with online hate speech leads to a tendency to produce hate speech [5], creating a contagious effect enhanced by the context of online environments.

In 2017, a study analysing YouTube video comments of former US President Donald Trump during his 2016 presidential campaign [5] showed a correlation between the offensiveness of an initial comment and the offensiveness of its responses. Another study[6] based on anti-Asian racist tweets at the time of the Covid epidemic showed that hateful tweets are about 5 times more retweeted. An initial comment with hateful language hence has a much higher probability to engender hate speech.

Similarly to fake news, in which contagiousness has been recognized, a hateful message online will generate more reactions, and will therefore also be more seen and recommended. Hateful behaviour is often more provocative, violent, and divisive; it generates negative emotions that are more likely to be reacted to than other emotions [7].

2.1.2 Social exclusion

All the information in this section is taken from a conversation with Mr.Laurent Rignault, online reputation specialist and CEO of SOS-e-reputation.com (annex 6.3).

e-reputation

Nowadays, reputation highly depends on online reputation, also called e-reputation. However, this new form of reputation is particularly fragile. Internet is a space that is particularly conducive to hate and its propagation. Therefore, there is a proliferation of cases of individuals or companies particularly targeted by online hate speech at an impressive speed and scale. This leads to the subject being excluded from society because of the online hate that has destroyed its e-reputation.

An ethical and democratic problem

In the democracies' *trias politica* model, the guiltiness of a person and his/her sentence is the role of the judiciary instances. However, with online hate speech, anyone can apply their law, decide to exclude someone from society, destroy his/her reputation, and create ethical and democratic problems.

The importance of not giving such a 'judiciary' power to anyone online can be seen in the phenomenon of "Revenge Porn". The principle: a person, to take revenge, posts online pornographic videos of someone online in the public sphere against their will. The victim suffers the consequences of a person who has applied his justice.

Moreover, it is practically impossible to change a bad e-reputation, negating the right to a second chance. Once a person has fulfilled the condemnation decided by a competent court, the right to reintegration into society is ethically crucial. However, Internet's tough memory makes it nearly impossible for an individual or a company to repair its online image.

However, this problematic is complex as freedom of speech online must be preserved and governments have to act against web international companies whose economic interest is to moderate as little as possible.

2.1.3 Violent consequences

Normalization and dehumanization

The proliferation of online hate normalizes it. On internet, anonymity and easy expression create a distance between the user and their speech, and a feeling of impunity online. This leads a deresponsibilisation of online speech, making online hate a normal event [8].

This normalisation of online hate is accentuated by effects such as the echo chamber. An echo chamber is an environment where an individual is confronted exclusively with his or her own opinion, which he or she will then validate through confirmation bias[9].

These Echo chambers and deresponsibilised hate cause a dehumanisation of the victim. Hannah Arendt has shown, in studying the case of the Nazi Eichmann, that the dehumanisation of the hated individual is the source of violence[10]. This can explain the violent consequences of online hate speech.

Online harassment and discrimination

Harassment is defined as a person or group acting systematically against another person or group against its will. To show how internet can amplify the violence of harassment, the case of cyberbullying (online school harassment) is explored here.

With cyberbullying, the victim continues to suffer the insults of his or her peers online even when back home. The continuity of the harassment encourages a devastating spiral; the victim responds more easily to his or her harassers until he or she becomes eventually convinced by them. Moreover, cyberbullying encourages the proliferation of insults and the number of bullies. Harassers can more easily go beyond the school, and it is common for a teenager to be harassed online by people who are not even in the school [11].

Online harassment, therefore, makes harassment even more violent, especially as it is combined with offline harassment to become permanent, with terrible consequences such as depression and suicide. The problem with online discrimination is similar, it completes offline discrimination and makes it permanent.

Discrimination is particularly present online. In 2021, the LGBTQ+ community was amongst the communities that experienced the most online hate, with a rate of 64% in the US [12]. In addition, Covid has been a catalyst for online discrimination. About 29% of victims of such injustices in the UK said that abuses had worsened during the health crisis[13].

Psychological consequences on victims

Anonymity is an important aspect of internet. On the one hand, anonymity allows people to keep their privacy, intimacy, and freedom. On the other hand, there are psychologically harmful aspects of anonymity. Indeed, with anonymity comes a feeling of impunity which incites hate. Within a group, anonymity favours polarisation and scapegoating[14].

A scapegoat is a minority person or group against whom a majority group unites. The victim is blamed for all the problems and suffers consequences. According to René Girard, the social psychology phenomenon of scapegoating is even more dangerous because it is unconscious and generates an exacerbated violence endemic to the majority group [15].

The consequences of online violence on the psychology of the victims are terrible. Online violence leads to a deterioration of mental health: loneliness, stress, low self-esteem... [16] Being a victim of cyberhate can lead to depression, which entails fatigue, sadness, loss of desire and pleasure, eating and sleeping disorders [17]... When depression worsens, it can lead to the suicide of the victim. In France, 20% of teenagers have suicidal thoughts [18].

From online hate to physical hate crime

A study conducted by Cardiff University in 2019 compares English police hate crime data with hate tweet data [19]. Researchers conclude that there is a strong correlation between them; a rise in hate tweets in one location comes with a rise in the number of hate crimes there.

This correlation doesn't necessarily entail a causal relationship. Nonetheless, the amount of online hate speech appears to be a good indicator of physical hate, which could be used as a tool to predict future crimes and identify rising tensions in a geographical area. Furthermore, online hate speech is an amplification factor of physical violence as it facilitates hate to spread, incitement to hate, and organization of hate crimes. Therefore, online hate speech amplifies, spreads, and incites physical violence.

2.1.4 Online hate speech and freedom of speech

Countries in the world have very different views of freedom of speech. Some countries adopt a very liberal but porous position on the subject, such as the USA. Other countries go as far as totally restricting the freedom of expression online. Finally, some countries try to keep freedom but limit hate speech, like France. In this part, we will debate different philosophical visions of freedom of speech while presupposing that freedom and the decrease of online hate speech are two valuable and desirable elements for our society.

An intuitive definition of freedom

Intuitively, freedom can be defined as the absence of constraint. This is how the philosopher Thomas Hobbes, for example, defined freedom in the course of his work [20]. When applied to online freedom of expression, it comes back to the American model without constraints stipulated in the first amendment [21].

However, this definition of freedom is very contested philosophically, considered to be wrong and dangerous. First, it is wrong because there are counterexamples. For example, in a dilemma, a person is not constrained but must choose between two unwanted options, which is poorly valuable freedom. Thus, Jean-Paul Sartre in *Existentialism is a Humanism* [22]. argues that freedom must be defined by the presence of a real choice of possibilities and not by the absence of constraints.

Secondly, this is a dangerous definition of freedom because it allows immoral acts. If I have no constraints then I can harm others and constrain the freedom of others. Not harming others is the moral limit of unconstrained freedom, and this limit is found at the heart of our principles, as in the 1979 *Declaration of the Rights of Man and of the Citizen* [23].

On the Internet, there is mainly freedom of speech with nearly no limit. The scale and immediacy of cyberspace mean that this massive freedom of expression, while fundamental, can be paradoxically toxic when infected with hate.

Freedom with limits

To avoid the dangers of unrestricted freedom, there should be a freedom which, in principle, stops at the harm of the other, meaning stopping at hate speech. To go further, there is the principle of moral freedom that Levinas develops in *Totality and Infinity* [24]. This principle puts the freedom of others before one's own to avoid the immorality of the arbitrary character of inequality of freedom. This principle allows counterbalancing possible deviance of restricted freedom by each person responsible for the other's freedom.

This constrained freedom is strong because it respects the principle of autonomy. The principle of autonomy consists in having freedom of possibility, as expressed by Sartre, with moral obligations. In other words, one can do what he/she wants if it is within the scope of moral obligations that he/she imposes on his/herself. This freedom, described by Heidegger for example in *The Essence of Freedom*, is valuable both in terms of freedom and morality [25].

It could be objected that it is not ourselves but national legislations that impose moral responsibilities, hindering the principle of autonomy. However, in a democracy, the principle of autonomy is respected at the scale of the population as the people vote for the legislation. Being a citizen in a democracy consists in accepting to impose on ourselves the laws decided by the population [26]. That's why models of freedom of speech by totalitarian regimes are not in consideration in our aim to favour freedom while fighting online hate speech.

Therefore, this philosophical section leads to a conclusion in favour of freedom with limits. Putting limits to freedom doesn't mean being less protective of freedoms; France is one of the countries with the most freedom of speech according to the Global State of Democracy Indices in 2020, ahead of the US [27].

2.1.5 Legislation of online hate speech

We will focus on two democratic areas with different visions of freedom of speech: the European Union (freedom with limits) and the USA (very liberal freedom).

Introduction to the issue

Most of the big digital platforms (Facebook, Twitter, Google...) are American platforms and follow liberal legislation regarding freedom of expression. This leaves us with a legal problem where EU states struggle to apply their laws online. Joël Rubinfeld, president of the LBCA (Belgian League Against Antisemitism), a specialist in online hate, has experienced this (source, see annex 6.3).

A Belgian Holocaust denialist Youtuber was in breach of the law since Belgian law prohibits Holocaust denial. However, his illegal content was available without restriction despite several requests from the LBCA. It was only when the LBCA made the case public, threatening to sue the platform, that Youtube banned the video in Belgium. This example shows the difficulties of applying legislation about freedom on web platforms.

Social Media policies

Most social networks have regulations that seem to restrict freedom concerning hate speech. For example, Twitter's policies state: "You may not promote violence against or directly attack or threaten other people on the basis of race, ethnicity, national origin, caste, sexual orientation, gender, gender identity, religious affiliation, age, disability, or serious illness" [28].

Despite this policy, online hate speech still grows, especially on Twitter, to the point that four NGOs sued Twitter in December 2021 in the French courts for lack of moderation of hateful content [29], leading to condemnation in January 2022 [30]. There are several reasons for this contrast between the policy of social platforms and the reality of online hate speech. For example, there is a technical limit as detecting online hate speech is a complicated task.

Beyond this technical limitation, what emerged from the discussions with Laurent Rignault and Joël Rubinfeld (annex 6.3) was a lack of clarity around platform policies, which platforms take advantage of to keep as much content as possible. As we saw with the example of the negationist YouTuber, if the hate speech is not publicly highlighted, it is difficult to get hateful content moderated.

Overview of American and European legislation

As mentioned earlier, the First Amendment of the US Constitution gives the right to absolute freedom of speech, and hate speech is therefore legal, including online. For example, in 1969, the US Supreme Court ruled that Clarence Brandenburg of the Ku Klux Klan was within his rights when he incited violence against minorities [31].

In most other democratic countries, there are laws limiting freedom of expression, which have recently been adapted to cyberspace. For example, in Germany, the Network Enforcement Act of 2017 provides for continuous monitoring of internet platforms to ensure the application of restrictions on freedom of expression, as well as the possibility of amending breaches [32]. In France, the Avia law, adopted in June 2020, marks a real milestone in the field: accountability of platforms, creation of an online hate observatory, 24-hour deadline to remove illegal hate content... [33]

With the European Union, a supranational legal framework is also emerging. This started in 2016 with the adoption of a code of conduct against online hate following negotiations with Facebook, Youtube, Microsoft and Twitter [34].

To impose its legislation, the European strategy is to economically sanction the platforms' infractions. This is the direction recently taken by the Digital Service Act, adopted on the 23rd of April 2022 [35]. To monitor compliance with the Digital Service Act, the major platforms will be pushed into full transparency, audited annually, and threatened with penalties of up to 5% of their daily turnover.

Limits to the European legislation

Putting in place supranational legislation is a long and complex process as it is exposed to confrontation with national laws and international companies. Therefore, this European legislation is not always effective and fast enough.

Moreover, the application of these constraints will always be very limited on the internet. If it is possible to negotiate with the big platforms, it will be very hard with all other websites to force them to respect the law. Telegram, for example, is a social network with many users and almost no moderation, allowing the expression of hate speech. Furthermore, these restrictions are European, therefore within a geographical limit, meaning a VPN would be enough to bypass them [36].

The future of the EU's agreements with the major digital platforms is also uncertain. 22nd April 2022, Elon Musk became the new owner of Twitter. He has expressed his intention to apply a libertarian and absolutist vision of freedom of expression, going in the opposite direction of the European Union. We can therefore expect a significant decrease in moderation on the platform and new tensions with the EU [37].

2.2 Automatic hate speech detection

2.2.1 Natural language processing

Definition and application

Natural language processing (NLP)[38, 39, 40] is a computer sciences domain making computers able to process human language. This field has seen huge improvements in the past decade including automatic machine translation, search engine and dialogue systems. These applications are based on several ideas including developing an algorithm, linguistics, and statistics. NLP can help solve important societal problems. For example, researchers have developed an NLP algorithm to detect the signals of a suicidal tendency and to assess the risk for a depressive person to commit suicide [41].

One of the first milestones in NLP was by Noam Chomsky in 1957 with *Syntactic Structures*, which suggested applying syntactic transformations to our language. This research led in 1966 to the first translation AI by the US National Research Council. In the 1980s, the research about NLP was fostered by new successful ideas based on the intuition to implement statistical models. This success continued throughout the 1990s, especially with the progress of machine learning and neural networks, leading in 2001 to the first neural language model. Nowadays, NLP is in our daily life with for example voice recognition AI assistants [42].

2.2.2 Machine learning

Definition

Machine learning [43, 44] is a branch of artificial intelligence which focuses on the use of data and algorithms to imitate the way that humans learn. Arthur Samuel coined the term machine learning in 1952 when he developed a model capable of playing checkers. Since then, board games, especially chess, have continued to be an area of research for the improvement of machine learning, with for example the memorable victory of Deep Blue over Kasparov in 1997. Since, advances in machine learning have multiplied exponentially, notably thanks to the implementation of neural networks and deep learning. Today, machine learning algorithms can very easily beat Deep Blue at chess [45].

Machine learning can be divided into three types of learning: Supervised Learning, Unsupervised Learning and Reinforcement Learning.

Supervised Learning [46]

With supervised learning, the algorithm learns from labelled data. For example, it is used in spamming recognition. An algorithm learns from a dataset with comments labelled as "spam" or not "spam". The model is then able to predict if a comment is spam or not. There are two main tasks in Supervised Learning:

- Classification models learn to assign data into specific categories or classes
- Regression models learn to understand the relationships between dependent and independent variables

The major issue of supervised learning is that it requires labelled data, which can be time consuming and expensive.

Unsupervised Learning [47]

Unsupervised learning is a type of machine learning that extracts information from unlabeled data. By analyzing patterns and structures, unsupervised learning can be used to do tasks like clustering, dimensionality reduction, text generation, etc. The advantage of this approach is that it doesn't require labelled data, but the result can sometimes be harder to understand from a human perspective. There also exists Semi-supervised learning, when only a small part of the data is labelled. It combines supervised and unsupervised learning to compensate for an eventual lack of labelled data.

Reinforcement learning [48]

Reinforcement learning is a type of machine learning based on reward. The model has an initial state and will attempt a path towards an outcome. It will get a certain amount of reward for this outcome. Its aim is to maximize its rewards, by attempting several paths. It is used for example in simulations to determine the best decision to take.

Deep learning[49, 50]

Deep learning is a branch of machine learning inspired by the structure of the brain which can be implemented in any of the three learning approaches above. It reproduces neural connections by using a multi-layered structure of algorithms called artificial neural networks. In recent years, deep learning achieved outstanding results in natural language processing and computer vision. However, due to the large number of parameters that must be learned, the training of deep neural networks requires a large database and a good computing capacity.

This work will focus on supervised learning for text classification. There are three steps in building such a model: prepossessing and vectorization, training a learning model, and evaluating this model.

2.2.3 Data Engineering

Preprocessing [40]

Preprocessing aims to only keep words with significant semantic value as NLP learning models are based on vocabulary. The first step of preprocessing is usually to simplify the text by lowering the text, removing special characters, punctuation etc.

To associate a weight to each word during the training, word frequency is commonly considered. Too rare or too common words usually have poor semantical importance. With the same idea, it is common to remove stop-words. Stop-words are very common words with poor semantic value, such as "I" "with" "for" "is" "which" etc.

Preprocessing reduces words to their semantic values, so that the words "writer", "written", "writing", "writes" are processed as the same word. There are two main methods to do so, lemmatization and stemming. Lemmatization consists in reducing a word to its lemma (its root), basing itself on a dictionary. "Is", "are", "were" etc. will be transformed to "be". Stemming keeps the root of the word by cutting its ending. Therefore, "consulting" and "consultant" would both become "consult".

Many other additional methods can be added. For example, disambiguation consists in using a dictionary to distinguish the different meanings of a word (such as a book and the verb to book).

Vectorization and tokenization [51]

Vectorisation consists in transforming texts into vectors of numerical values that can be used during the training. There are two main strategies of vectorizations.

The first one is called bag of words. In this approach, the input text is used as an unordered list of words. Each word (or token) is assigned a numerical id. To improve this method, it can be added a counting element, making it a count vectorizer. This counts the frequency of the tokens to give them a weight. A counting vectorizer can be complexified into a tf-idf (Term Frequency–Inverse Document Frequency). Tf-idf takes in account the importance of each word in the context of its sentence and gives it a weight. It calculates the weight of a word in a sentence with the following formula:

f(w)=frequency of the word in the sentence
t(w)=total number of words in the sentence
t(d)=total number of sentences in the dataset.

t(s)=total number of sentences containing the word.

$$tf - idf = \frac{f(w)}{t(w)} * log(\frac{t(d)}{t(s)}) \tag{1}$$

The idea is to give more importance to very frequent words within a sentence, but less importance if these words are very frequent within the whole dataset. A token and its associated weight constitute a feature used during the training.

To keep some sense of word order within a text data, it can be adopted an n-gram strategy, consisting in giving weights not to each word, but to each group of consecutive words (the size of the group being determined by an n value).

The second strategy is word embedding [52]. The idea is to keep the semantic, the syntactic and the contextual values of a word by transforming each word into a representative vector. Each sentence becomes a vector of vectors. Word embedding uses neural networks to cluster words with similar syntactic and semantic values and creates relations between them. From this process results a vector to which each word is associated too. This strategy is based on the distributional hypothesis which states that words with similar meanings will be close to each other in their vector representation. By taking into account the relations between words, word embedding captures better the meaning of each word.

2.2.4 Some learning algorithms

The following learning techniques have been used and compared in this dissertation:

Decision tree classifier[53] is a type of supervised learning method with a tree-like structure used for classification and regression. It is essentially composed of three major components: root, branches, and nodes. Each node corresponds to a decision applied to the data. Branches indicate the type of decision that has been taken, and final nodes correspond to the class predicted by the algorithm. They are easy to understand and can be applied with numerical and categorical values. The depth of the decision tree can be varied to optimize performance. However, they can be unstable from one training to the other and can easily overfit the data. Overfitting is a problem consisting of a model fitting its training data to the point it provides poor generalization to new data.

Random forest classifier[54] is an ensemble method used to reduce the variance obtained with a single decision tree. The principle of the method is to average the decision taken by many decision trees. Each tree of the forest is generated with a modified learning algorithm, where each node split is associated with a random set of features. The average decision over a large number of trees is used to decrease the variance and the risk of overfitting. The number of trees and their depths can be varied to optimize performance.

Logistic regression[55] is a binomial classification algorithm based on the relationship between input features and the probability of a particular outcome. This method is based on the logistic function (sigmoid) to model the binary outputs.

Support vector classifier(SVC)[56] is a model whose principle is to find the optimal hyperplane that separates the data. The optimization is done by maximizing a marge between the classes and the hyperplane. The two classes must be linearly separable, but some techniques called kernel functions can be applied to the data so that the classes become linearly separable [57]. The kernels can be varied to optimize performance. In this dissertation the linear kernel, the sigmoïd kernel and the Gaussian radial basis function (RBF) were used and compared.

K-nearest neighbours[58] is a type of supervised learning algorithm used for both regression and classification. The algorithm predicts the correct class for the test data by calculating the distance between the test data and all the training points. It then selects the K number of points which are the closest to the evaluated point and the class with the majority of points is selected. The number of 'neighbours' can be varied to optimize performance.

Naive Bayes[59] is a classifier based on Bayes' probabilistic equation2. This equation stipulates that taking events A and B, the probability of A knowing B is equal to the probability of A multiplied by the probability of B knowing A divided by the probability of B.

$$P(A|B) = P(A)\frac{P(B|A)}{P(B)} \tag{2}$$

During the training part, the model will use this equation to calculate the probabilities of the features knowing to which class the data entry belongs. Then, it will use these probabilities to make a classification prediction. To apply this equation, the 'naive' assumption that the features are conditionally independent is made. Several distribution methods of the Naive Bayes algorithm were tested in this dissertation: Multinomial method, Complement method, and Bernoulli method.

2.2.5 Evaluating a model

In binary classification, when applying the trained model to a test dataset, four elements are measured:

- True positive (TP): Number of points correctly predicted as class 1
- True negative (TN): Number of points correctly predicted as class 0
- False positive (FP): Number of points falsely predicted as class 1
- False negative (FN): Number of points falsely predicted as class 0

These amounts will allow us to calculate different evaluating metrics.

Confusion matrix [60]

The confusion matrix is a conventional way to plot the results of a model. It allows to see intuitively the performance of the model and understand the results behind the various metrics. Figure 1 is an example.

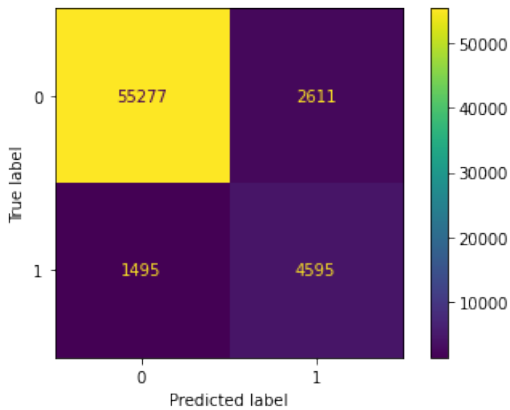

Figure 1: Confusion Matrix example for a binary classification

The top left value corresponds to true negatives, the top right value corresponds to false positives, the bottom left value corresponds to false negatives, and the bottom right value corresponds to true positives. Therefore, we want to maximise the top left / bottom right diagonal and minimize the top right / bottom left diagonal.

Accuracy [61]

The accuracy corresponds to the ratio of correct predictions over the total amount of predictions (Eq 3). It is the most common metric, but in a very unbalanced dataset (meaning one class is much more represented than others), the accuracy is a misleading metric. In a dataset with 90% negative entries and 10% positive entries, a model saying that 100% of the data is negative will be an inefficient model but with an accuracy of 90%.

$$accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{3}$$

Balanced accuracy [62]

This metric is a modified accuracy for unbalanced dataset. It introduces weights to counter the defaults of the accuracy on this type of dataset (Eq 4).

$$b_acc = \frac{1}{2} * \frac{TP}{TP + FN} + \frac{1}{2} * \frac{TN}{TN + FP} \tag{4}$$

Precision, recall and f1 score [63, 64]

Precision is a metric corresponding to the ratio of true positive predictions amongst all positive predictions. Recall is a metric corresponding to the ratio of true positive predictions amongst all positive data. Precision measures the ability to not be wrong when predicting positively (Eq 5). Recall measures the ability of the model not to miss positive values (Eq 6).

The f1 score is a synthesis metric of precision and recall, which are very complementary. It corresponds to a harmonic mean of these two metrics (Eq 7). The use of a harmonic mean makes it particularly efficient to evaluate models tested on unbalanced datasets.

$$precision = \frac{TP}{TP + FP} \tag{5}$$

$$recall = \frac{TP}{TP + FN} \tag{6}$$

$$f1_{score} = \frac{2}{(1/precision) + (1/recall)} \tag{7}$$

2.2.6 Current situation

Nowadays, implementing neural networks, and thus deep learning, gives the best results. However, deep learning takes much more time and requires much more data as it creates itself the features used for training. It determines the optimal combination of functions, techniques, and parameters for the task it has been given [65].

In 2018, a breakthrough was made concerning NLP with the development of a new language model called BERT (Bidirectional Encoder Representations from Transformers)[66]. Applying bidirectional training to language modelling appears to be a method much more able to consider the context of a word when vectorizing it, contrarily to a classic word embedding method. Nowadays, BERT is to be the most performant approach concerning NLP, notably for automatic hate speech detection [67].

3 Experiment

The growing and dangerous problem of online hate, described in 2.1, raises the question of its detection. Beyond the legal aspect, the technical implementation is a huge challenge. The amount of content is too important for manual moderation, so there must be an automatic selection. Therefore, platforms are developing automatic hate speech detection tools, algorithms capable of detecting online hate. In addition to combating the harmful consequences of online hate, the challenge of effective hate speech detection is also to stem its spread as quickly as possible and mitigate the effect of its contagiousness.

In this experiment, I develop a tool capable of detecting 6 classes of hate. Creating 6 independent binary classifiers models. This tool is then applied in several use cases for more analysis about hate speech. We use Python 3.7.13 and a Google Colab notebook environment. The two main libraries used are NLTK, a library specialized in NLP, and SciKit Learn, a library specialized in machine learning. The code is entirely available through an URL in annex 6.2.

3.1 Dataset

3.1.1 Dataset presentation

To train our model, I used a public labelled dataset from Kaggle [68] (annex 6.1) that classifies Wikipedia comments to 6 classes: toxic, severe toxic, obscene, threat, insult, and identity hate. Each comment has been manually labelled and can belong to several classes. Obscene, threat and insult are self-explanatory classes.

- Toxic: a harmful comment that could initiate more online hate.
- Severe toxic: a particularly harmful and potentially contagious hateful comment.
- Identity hate: hate content refering to an identity element (race, religion, gender, sexuality, origin etc.).

This dataset has many advantages: it is labelled, public, and easily exploitable. Its main drawback is that it is very unbalanced, and some classes lack data.

The CC0 and CC BY-SA 3.0 licenses of the dataset allow us to use it without any propriety or ethical issue.

3.1.2 Data description

As described in Table 1, 224k comments have been used for the study. The dataset has been randomly separated into training and test sets. They respectively correspond to 71% and 29% of the full dataset.

	Total	Training	Test
N comments	223549	159571	63978
%	100	71	29
average length (characters)	391	394	383

Table 1: Description of the dataset

Tables 2 and 3 respectively show the number and percentage of each comment per class in the training and test sets. As we can see, the classes are highly imbalanced and classes like threat or identity hate only include 478 and 1405 comments of the training set. For these classes, we can expect larger statistical uncertainties and unstable training. The training and testing datasets both have similar class distribution. The any hate class was added manually as the sum of the six hate classes.

Class	toxic	severe toxic	obscene	threat	insult	identity hate	any hate	no hate
N	15294	1595	8449	478	7877	1405	35098	124473
%	9.6	1	5.3	0.3	4.9	0.9	22	78
average length (characters)	295	454	287	308	277	308	303	404

Table 2: Description of the training dataset - class statistics

Class	toxic	severe toxic	obscene	threat	insult	identity hate	any hate	no hate
N	6090	367	3691	211	3427	712	14498	49480
%	9.5	0.6	5.8	0.3	5.4	1.1	22.7	77.3
average length (characters)	244	731	268	338	261	326	243	398

Table 3: Description of the test dataset - class statistics

Representativeness is a metric of how much one class is represented in another. For example, 10% of the toxic class is represented in the severe toxic one, making the severe toxic class not very representative of the toxic. Reading Table 4, more than 90% of each class is represented in the toxic class, the toxic class is by far the most representative one of the datasets.

Example on how to read Table 4: the value in 'toxic' column, 'severe toxic' row is 10.4. It means that 10,4% of the toxic data are also labelled severe toxic. The value of 'severe toxic' column, 'toxic' row is 100%, meaning 100% of the severe toxic data are also toxic data.

%	toxic	severe toxic	obscene	threat	insult	identity hate
toxic	**100.0**	10.4	51.8	2.9	48.0	8.5
severe toxic	100.0	**100.0**	95.1	7.0	86.0	19.6
obscene	93.8	18.0	**100.0**	3.5	72.8	12.2
threat	93.9	23.4	63.0	**100.0**	64.2	20.5
insult	93.2	17.4	78.1	3.9	**100.0**	14.7
identity hate	92.7	22.2	73.5	7.0	82.6	**100.0**

Table 4: Description of the training dataset - class representation

Correlation is a numerical value indicating the link between classes. A correlation of 0 means the two variables are independent, while a correlation of 1 means they are fully dependent. As illustrated in Figure 2, toxic, obscene and insult seems quite related to each other.

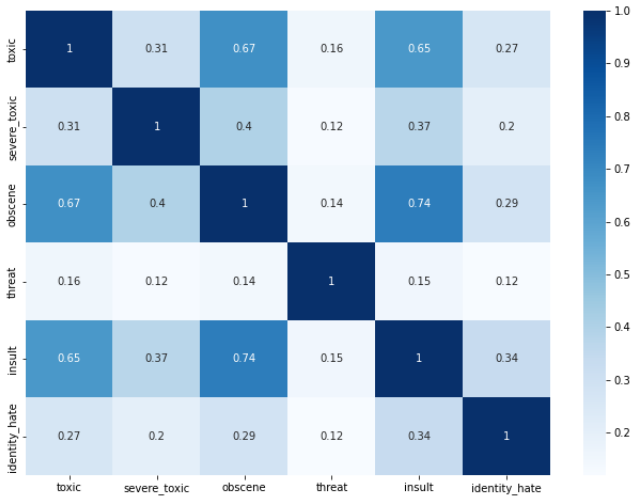

Figure 2: Description of the training dataset - class correlation

3.2 Methodology

3.2.1 Data preparation

Data cleaning

A lot of the comments from the test files were not labelled, and thus unusable. It required some data cleaning. The statistics in Table 1 describe the datasets after cleaning.

Preprocessing

The text data was preprocessed using the techniques described in section 2.2.3. I started by putting all the text in lower case, then by removing links, URLs and emails. Afterwards, I removed the punctuation and all the non-English characters. I removed very short words, very long words and stop words. I then applied lemmatization and stemming to each word. Finally, I use a word frequency threshold to remove all the words which appear fewer times than this threshold, to only keep significant vocabulary. This preprocessing was optimized in part 3.3.2.

Vectorization

Afterwards, the text data was transformed into numerical values exploitable by the algorithm using two vectorization methods, counting vectorization and tf-idf, as previously described in section 2.2.3.

Here is an example from the training dataset:

- After data cleaning: "They are NOT ORIGINAL research, they are pointed in the episodes."
- After preprocessing: "origin research point episod"
- After tf-idf vectorization (word id, weight of the word): (421, 0.7067810013503653), (302, 0.4074120310468896), (1289, 0.2782526865946295), (39, 0.507002461030806).

3.2.2 Training 6 independent binary classifiers

Choice of the toxic class

Due to time and computation constraints, only one class has been used to compare the different learning techniques and their hyperparameters. Once an optimized configuration was obtained, it was used to create train 6 detection models, one per class.

The toxic class has been chosen to determine an optimal configuration. It is the largest class in the dataset and the most representative one.

Model selection

I have trained the different learning methods described in 2.2.4 with various hyperparameters to determine the best configuration for our task. Five metrics described in 2.2.5 have been compared: accuracy, balanced accuracy, f1 score, precision, and recall. As the dataset is very unbalanced, the focus has been set on balanced accuracy and f1 score metrics.

Model optimization

Several parameters have been varied to find the optimal setting:

- The preprocessing between a lemmatization process, a stemming process, both of them or none.
- The word frequency threshold, designing the frequency threshold under which a word is removed during the preprocessing.
- The vectorization method between a count vectorization and a tf-idf vectorization.
- The type of ngram used for the feature creation

Once an optimized model is found for the toxic class, its configuration is used to train the other classifiers in order to have 6 detection models, one for each type of hate.

3.3 Results

3.3.1 Learning Model Selection

Initial parameters The learning models have been tested with these initial parameters, optimized in 3.3.2:

- Preprocessing with both lemmatization and stemming applied
- word frequency threshold = 5
- Vectorization: tf-idf, unigrams

SVC

The SVC learning algorithm was trained then tested, varying the kernel. The results are shown in Table 5. The three kernels obtained similar scores: around 85% balanced accuracy and 68% f1 score. The f1 score is lower than the balanced accuracy due to a precision of around 60%, meaning an important ratio of data was falsely considered toxic. SVC learning models were the longest ones to train due to their sensibility to the high number of features, especially with an RBF kernel. Considering training time and balanced accuracy, the linear kernel seems to be the one that learned the most.

Model	kernel	acc	*b_acc*	f1	precision	recall
SVC	linear	0.931	0.867	0.685	0.606	0.788
SVC	sigmoid	0.926	0.862	0.668	0.582	0.782
SVC	rbf	0.936	0.855	0.691	0.638	0.755

Table 5: Comment classification task scores for the SVC model, varying kernel - toxic class

K nearest neighbour

The K nearest neighbours learning algorithm was trained then tested, varying the number of neighbours. The results are shown in Table 6. The f1 score being under 50% for every KNN means this algorithm did not manage to learn how to detect toxicity. Particularly, the recall metric is very low, meaning the models failed to detect toxic data correctly.

Model	k	acc	*b_acc*	f1	precision	recall
kNN	2	0.919	0.636	0.402	0.68	0.286
kNN	3	0.917	0.651	0.426	0.627	0.323
kNN	4	0.918	0.635	0.399	0.666	0.285
kNN	5	0.917	0.645	0.414	0.63	0.308
kNN	6	0.919	0.631	0.393	0.684	0.275
kNN	7	0.918	0.635	0.399	0.666	0.284
kNN	8	0.919	0.623	0.377	0.699	0.258
kNN	9	0.918	0.626	0.382	0.684	0.265
kNN	10	0.918	0.617	0.364	0.707	0.245

Table 6: Comment classification task scores for the KNN model, varying k - toxic class

Decision tree

The decision tree learning algorithm was trained then tested, varying the depth of the tree. The results are shown in Table 7. This algorithm scored well on balanced accuracy but not that much on the f1 score due to low precision, like for the SVC models.

Model	depth	acc	*b_acc*	f1	precision	recall
dTree	5	0.921	0.692	0.497	0.635	0.409
dTree	10	0.927	0.749	0.58	0.642	0.529
dTree	20	0.924	0.791	0.611	0.595	0.628
dTree	30	0.926	0.809	0.63	0.599	0.664
dTree	40	0.922	0.819	0.629	0.577	0.691
dTree	50	0.921	0.827	0.631	0.567	0.712
dTree	60	0.918	0.827	0.625	0.556	0.713
dTree	70	0.919	0.834	0.631	0.557	0.728
dTree	80	0.916	0.836	0.627	0.545	0.737
dTree	90	0.916	0.845	0.631	0.541	0.757
dTree	200	0.909	0.85	0.619	0.514	0.777
dTree	500	0.905	0.853	0.613	0.502	0.789

Table 7: Comment classification task scores for the decision tree model, varying depth - toxic class

Random forest

The random forest learning algorithm was trained then tested, varying the depth and number of the trees. The results are shown in Table 8. The random forest models do perform better as the number of trees and their depths increases, however, it seems to reach a plateau from 100 trees of 300 depth, resulting in 85% balanced accuracy and 68%. A larger amount of trees and a higher depth increase the time of training. Considering this, the model with 100 trees of 500 depth is the one to prefer amongst the random forest models.

Model	N	depth	acc	*b_acc*	f1	precision	recall
RF	100	10	0.905	0.5	0.0	1.0	0.0
RF	100	100	0.936	0.698	0.545	0.832	0.406
RF	100	200	0.94	0.826	0.686	0.687	0.685
RF	100	300	0.936	0.845	0.687	0.646	0.733
RF	100	400	0.934	0.849	0.682	0.629	0.744
RF	**100**	**500**	**0.933**	**0.855**	**0.683**	**0.621**	**0.76**
RF	150	500	0.933	0.856	0.683	0.62	0.76
RF	150	600	0.933	0.857	0.684	0.619	0.764
RF	150	700	0.932	0.857	0.682	0.615	0.764
RF	200	400	0.935	0.851	0.685	0.633	0.747
RF	200	300	0.937	0.847	0.689	0.647	0.737
RF	200	500	0.933	0.855	0.684	0.623	0.758
RF	200	600	0.934	0.857	0.686	0.623	0.763
RF	200	700	0.932	0.857	0.683	0.617	0.764
RF	250	600	0.933	0.858	0.685	0.62	0.764
RF	250	700	0.932	0.86	0.685	0.616	0.771
RF	300	500	0.933	0.854	0.682	0.621	0.756

Table 8: Comment classification task scores for the random forest model, varying depth and N trees - toxic class

Logistic Regression

The logistic regression learning algorithm was trained then tested, varying the norm. The results are shown in Table 9. With an l2 norm, the logistic regression learning method did better in terms of f1 score. The results are quite similar to the SVC models, with similar interpretations to be made.

Model	Norm	acc	*b_acc*	f1	precision	recall
logistic regression	None	0.917	0.855	0.642	0.547	0.777
logistic regression	L2	0.936	0.839	0.681	0.648	0.718

Table 9: Comment classification task scores for the logistic regression model, varying norm - toxic class

Naive Bayes

The Naive Bayes learning algorithm was trained then tested, varying the distribution method. The results are shown in Table 10. The different learning methods failed to reach a convincing f1 score (always under 60%), for different reasons. The multinomial learning method failed to learn to recognize correctly toxic data. The complement and the Bernoulli methods often labelled non-toxic data as toxic.

Model	Norm	acc	*b_acc*	f1	precision	recall
NB	Multinomial	0.935	0.734	0.586	0.736	0.487
NB	Complement	0.848	0.86	0.522	0.372	0.875
NB	Bernoulli	0.664	0.778	0.343	0.21	0.92

Table 10: Comment classification task score for the Naive Bayes model, varying learning distribution method - toxic class

Conclusion

The linear SVC model and the random forest with 100 trees of 500 depth model are the ones that performed the best, with around 85% balanced accuracy and 68% f1 score as shown in Table 11. However, considering that the linear SVC algorithm was about 3 times longer to train, the random forest has been used for the rest of this dissertation.

Model	acc	*b_acc*	f1	precision	recall
LR [L2]	0.936	0.839	0.681	0.648	0.718
RF [N100, d500]	**0.933**	**0.855**	**0.683**	**0.621**	**0.76**
SVC [linear]	0.931	0.867	0.685	0.606	0.788

Table 11: Best selected models for the tweet classification task - toxic class

3.3.2 Optimization

Optimizing preprocessing

The random forest model has been trained with varying preprocessing (lemmatization only, stemming only, both or none). However, no significant difference has been noticed in the results. As applying both stemming and lemmatization reduces the time to create features, this preprocessing method has been chosen to proceed.

Optimizing word frequency threshold

The random forest model has been trained with varying the word frequency threshold, and then tested as shown in Table 12. The balanced accuracy and the f1 score follow opposite curves as the threshold increases, due to a growing recall and a decreasing precision. The threshold of 5 that maximises the f1 score is chosen to proceed.

Threshold	acc	*b_acc*	f1	precision	recall
2	0.935	0.841	0.681	0.643	0.726
5	**0.933**	**0.855**	**0.683**	**0.621**	**0.76**
6	0.93	0.857	0.678	0.609	0.766
8	0.93	0.863	0.679	0.602	0.78
10	0.929	0.864	0.678	0.598	0.784
20	0.927	0.868	0.675	0.587	0.796

Table 12: Performance as a function of the word frequency threshold - random forest with 100 trees of 500 depth - toxic class

Optimizing vectorization method

The random forest model has been trained varying the vectorization method, then tested as shown in Table 13. Tf-idf gives slightly better results than the counting vectorizer. This is not surprising considering tf-idf is an improved version of count vectorization that takes more information into account while creating features.

Vectorization	acc	*b_acc*	f1	precision	recall
Count Vectorization	0.928	0.852	0.668	0.593	0.764
Tf-Idf	**0.933**	**0.855**	**0.683**	**0.621**	**0.76**

Table 13: Performance as a function of the vectorization method - random forest with 100 trees of 500 depth - toxic class

The random forest model has been tested varying n-grams during the tf-idf vectorization, then tested as shown in Table 14. Considering only unigrams (one word=one feature) is the best option according to our results. Interestingly, considering keeping bigrams or trigrams brings a very high precision but a very low recall, meaning the model rarely gives false negatives but did not learn to distinguish a toxic and a non-toxic data.

n	acc	*b_acc*	f1	precision	recall
1	**0.933**	**0.855**	**0.683**	**0.621**	**0.76**
1/2	0.928	0.635	0.419	0.909	0.272
2	0.911	0.535	0.131	0.867	0.071
1/2/3	0.918	0.572	0.251	0.949	0.144
2/3	0.907	0.516	0.063	0.847	0.033
3	0.905	0.501	0.003	0.889	0.001

Table 14: Performance as a function of the n-grams parameter for vectorization - random forest with 100 trees of 500 depth - toxic class

In conclusion, random forest with 100 trees of 500 depth with a word frequency threshold of 5 and a tf-idf vectorization is the optimal setting in our case.

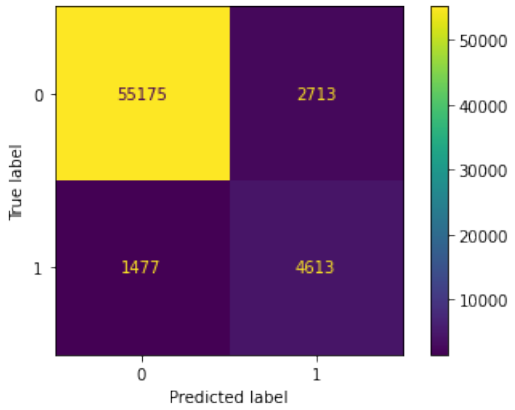

Figure 3: Confusion Matrix- random forest with 100 trees of 500 depth - toxic class

3.3.3 Adapting the model to each class

The optimal setting found in sections 3.3.2 and 3.3.1 was then used to train classifiers for each hate class. These classifiers were tested as shown in Table 15.

The obscene class and the insult class got very similar performance results to the toxic class. However, the results for severe toxic, threat and identity hate are not satisfactory at all, with very low recalls (and therefore f1 score). This is due to the fact there are only very few data for these classes; 1595 entries (1% of dataset) for severe toxic, 478 entries (0.3% of dataset) for threat, and 1405 entries (0.88% of dataset) for identity hate. This amount of data was insufficient for our model to learn.

To not leave unused the data of these three classes, it was added an "any hate" class, which concerns all the data of the training dataset labelled with at least one of the other classes. The any hate class model has very similar results to the toxic class and did learn at least partially to detect hate speech as shown in Table 15.

Class	acc	*b_acc*	f1	precision	recall
Toxic	0.933	0.858	0.685	0.619	0.766
Severe Toxic	0.993	0.527	0.085	0.198	0.054
Obscene	0.96	0.83	0.665	0.649	0.682
Threat	0.997	0.512	0.045	0.455	0.024
Insult	0.958	0.773	0.592	0.622	0.565
Identity Hate	0.99	0.587	0.281	0.721	0.174
Any	0.929	0.865	0.684	0.607	0.784

Table 15: Performance of the binary classifier models for each class - random forest with 100 trees of 500 depth

3.4 Discussion

3.4.1 Comments on our detection tool

SVC and random forest algorithms gave the best predictions and managed to learn at least partially to detect hate speech. In the end, taking also into account the training time, we found random forest to be the most appropriate algorithm for this task.

For three of our classes, too little data was available to achieve exploitable models. These three classes have therefore not been used in the use cases.

While we achieved good scores in terms of balanced accuracy, our model had a lower f1 score than balanced accuracy due to a non-negligible amount of false positives. However, in this case, it is better to have false positives than false negatives. Indeed, it is better that no hateful content is forgotten by the algorithm, even if a manual moderation is needed afterwards to correct the false positives.

3.4.2 Possible improvements

Our training was optimized using the toxic class. A more rigorous methodology would have been to test all the learning classifiers and all the possible parameters/optimizations on all the classes. We could then have different configurations for each class, optimizing the global performance.

Then, it would be interesting to implement neural networks systems to do deep learning, which generally gives better results. Although this is more complicated to implement, it allows it allows for automation and improves the optimization of the model. Similarly, a word embedding vectorization could be tested. These techniques, described in 2.2, make the algorithm take better into account the context of a word. Eventually, this difficulty in understanding a word in its context led our model to have such an amount of false positives, relying too much on the meaning of a word in itself and not within a sentence.

Finally, the dataset could be extended by incorporating data from other social networks. We have considered that as Wikipedia comments are of the same length and nature as on other online platforms in general, it was sufficient to have an exploitable model, but having a more varied dataset would help fight overfitting.

3.4.3 Future challenges of hate speech detection

Current methods rely heavily on semantics, slightly on syntax and context, but still fail to perceive the intention of a sentence, which sometimes involves grammar for example. Thus, elements such as sentiment analysis detection or word disambiguation, although already developed, still have an important progress margin before the reaching subtlety of human language [69].

The two main current challenges of automatic hate speech detection are fighting overfitting and biases. These risks are usually linked to the presence of very frequent -but not representative- words or the presence of several time the same users in the training dataset. Furthermore, this overfitting problem can result in dangerous biases when the dataset is not representative enough of reality. Some models built by researchers have been proven to contain gender and/or racial biases[65].

4 Use cases

The second objective of this dissertation is to use our model distinguishing three types of hate as an analytical tool. For this purpose, three use cases were chosen about politics and coronavirus, two subjects particularly linked to hate speech (see 2.1.1).

These use cases were based on public datasets found on Kaggle, all with a CC0 public domain licence. As Twitter is a very large platform where expression is particularly prolific and instantaneous, it is a good platform on which to conduct online hate speech analysis.

NB: The % in the results Tables are the % within one class. For example, 0.47% of Toxic in the Stay column in Table 16 means that 0.47% of the 'Stay' class tweets are toxic.

4.1 Random tweets

First, we need statistics on the level of hate speech on Twitter in general. I therefore applied our model to a database of 20050 randomly selected users (predictions are in Table 16). For each user, a random tweet was selected. By taking a large random sample of tweets, this gives us a baseline of hate speech on Twitter (annex 6.1)[70].

Class	N	%
Toxic	1345	6.71
Obscene	833	4.15
Insult	473	2.36
Any	1391	6.94

Table 16: Predictions of the binary classifier model - random dataset - random forest 100 trees 500 depth

4.2 Brexit tweets

Over the past decade, a strong rise in nationalism in Europe led the UK in 2016 to vote to leave the EU in a referendum. Brexit is a representative example of the current cleavages and tensions in Europe. We study here the online generation of hate by politicians around this debate [71].

4.2.1 Dataset

This dataset collects over 53,000 tweets on the Brexit by the 449 MPs in post at the time, ranging from the early days of Brexit to the date of the vote in June 2016 (annex 6.1). These tweets are labelled according to the position of the MP who posted them. Looking at its description Table 17, it can be noticed that 3/4 of the MPs were against the Brexit but that they represent only 2/3 of the tweets, showing a stronger engagement of the pro-Brexit MPs [72].

	Total	Stay	Leave
Tweets	53576	35374	18202
Tweets %	100	66.03	33.97
MPs	449	340	109
MPs %	100	75.72	24.28

Table 17: Description of the Brexit tweets dataset

4.2.2 Results and Analysis

The predictions of our models are in Table 18. The number of hateful tweets is particularly low. This can be explained by the very polished language induced by such an important public function. It is also particularly non-obscene compared to what is generally observed on Twitter.

However, MPs are among the main actors in politics, which is the theme that generates the most hate. Thus, it is potentially not the language but subtle turns of phrase and divisive ideas of public figures that will subsequently generate hate speech.

Pro-Brexit MPs produced proportionally twice as many hateful tweets as anti-Brexit MPs; however, two limitations make it difficult to consider this result representative. Firstly, the very low number of hateful tweets in each category. Secondly, the fact that the number of tweets per MP is potentially very uneven.

Class	Stay		Leave	
	N	%	N	%
Toxic	167	0.47	171	0.94
Obscene	14	0.04	21	0.12
Insult	25	0.07	40	0.22
Any	224	0.63	210	1.15

Table 18: Predictions of the binary classifier model - Brexit dataset - random forest 100 trees 500 depth

4.3 Political tweets

The subject of hate speech revolves around the philosophical debate of freedom of expression and its legislative application. In this debate, the European view is opposed to the American one. In this use case, we will therefore compare hate speech tweets with #politics according to their location: USA or EU.

It is interesting to also compare the American tweets with some other English-speaking nationalities. Australia, UK, and Canada are democratic English-speaking nations comparable to the US and Europe culturally. In these three countries, like in the EU, freedom of speech is guaranteed by constitutional law with some legal limits [73] [74][75]. However, Canada and UK have a more libertarian version of freedom of speech than the EU. For example, in the UK, incitement to racial hatred is illegal, but Holocaust denial is not. These nations therefore form an interesting middle ground between the EU and the US.

4.3.1 Dataset

This dataset includes 198955 tweets collected on #politics between 24 July 2021 and 3 April 2022 (see its description in Table 19). More than 12.000 locations have been divided into 5 geographical areas and a final 'other' class (annex 6.1) [76].

	Total	US	UK	Australia	Canada	EU	Other
N	198955	47427	6305	2420	1996	2963	136848
%	100	23.84	3.17	1.27	1	1.49	68.78

Table 19: Description of the #Politics tweets dataset

4.3.2 Results and Analysis

The predictions of our models are in Table 20. Surprisingly, even though the tweets from the politics category generated more hate than those from the English MPs, the percentages are still much lower (around 2% for the any hate class) than those from our random tweet dataset (almost 7% for the any hate class). We can hypothesise that if politics is the theme generating the most hate online, it is more related to partisan tweets than a general hashtag on political comments. Notably, the tweets in our dataset are particularly non-obscene compared to what is observed in general on Twitter.

The 'other' class, by its size and plurality, gives us a general idea of the hate on twitter's #politics. It is interesting to note that Australia (1.32% toxic) and the EU (0.98% toxic) have fewer hateful tweets than the 'other' class, while the US (2.4% toxic), UK (2.17% toxic) and Canada (3.66% toxic) are above. This splits our geographical groups in two; Australia and the EU produced significantly fewer hateful tweets. They are also less libertarian on freedom of speech than the three other classes.

While this may show the beginnings of a correlation, we cannot yet assert causality between a country's legislation and the rate of online hate speech. This is due to some limitations of our use case. Firstly, only two EU countries (Malta and Ireland) have English as an official language. Thus, most of the EU tweets are from people who do not tweet in their language, which creates a significant bias. Furthermore, this correlation could be explained by a cultural difference and not a legislative one, although the two are linked. Finally, the strength of this correlation is hindered by the fact that Canada has produced significantly more hate than the US, despite a less libertarian policy.

Indeed, Canada produced more hateful (4.26%) and more obscene (1.3%) than other countries. My hypothesis is that the September 2021 federal election in Canada has created more political tension than in other countries, favouring hateful tweets, during the period covered by the dataset [77]. Other countries, such as

France, have also recently experienced tense elections during this period, but for France there is this language bias.

Class	US		UK		Australia		Canada		EU		Other	
	N	%	N	%	N	%	N	%	N	%	N	%
Toxic	1146	2.4	137	2.17	32	1.32	73	3.66	29	0.98	2833	2.07
Obscene	242	0.51	37	0.59	17	0.70	26	1.3	4	0.13	1071	0.78
Insult	217	0.46	26	0.41	7	0.29	9	0.45	7	0.24	666	0.48
Any	1290	2.72	150	2.38	37	1.53	85	4.26	32	1.08	3192	2.34

Table 20: Predictions of the binary classifier model - politics dataset - random forest 100 trees 500 depth

4.4 Coronavirus tweets

The health crisis of recent years is a topic that has generated a lot of tension in the world, especially politically. It is therefore interesting to conduct a study like that of #politics but with Covid-related tweets. The same geographical categories are used as classes: USA, UK, Australia, EU and Canada (annex 6.1).

4.4.1 Dataset

The dataset collected all tweets with hashtags related to covid during the period between the of 16th April 2020 and the 22nd of April 2020 included. The tweets were labelled with country codes, which were distributed between five geographical classes: USA, UK, Australia, EU and Canada [78]. The dataset is described in Table 21. Due to a large number of tweets (about 2.7 million entries), the automatic hate speech detection tool was only applied to the tweets corresponding to one of the 5 classes. Therefore, for our model to be able to process the amount of data, no 'other' class was created this time. Also, a data cleaning to only keep English tweets was applied.

	US	UK	Australia	Canada	EU	Total
N	30908	11385	1294	4194	4268	52 791

Table 21: Description of the coronavirus tweets dataset

4.4.2 Results and Analysis

The predictions of our models are in Table 22. First, although the proportions of hateful tweets are higher than in the first two use cases, they are still surprisingly below the norm determined by our random dataset. Nevertheless, if we compare them to the results of the Politic dataset, we can conclude that the tweets on the

covid were written in a less polished language and closer to the reality of the social network. In particular, the number of obscene tweets increased sharply in proportion to the other classes.

Interestingly, these results are in line with those of #Politics regarding a possible correlation between legislation and online hate. Indeed, we could divide the results into three groups; the EU with 2.25% of hateful tweets; the UK, Canada, and Australia between 3.19 and 3.32%; and the USA with 4.96%. These results point to a correlation between legislated freedom of expression with limits and a decrease in online hate, as the more libertarian the legislation, the more hateful were the tweets. However, the hypothesis of causality and the strength of this correlation are again hindered by a possible language bias or by cultural explanations for these results.

Class	US		UK		Australia		Canada		EU	
	N	%	N	%	N	%	N	%	N	%
Toxic	1404	4.54	337	2.96	38	2.94	130	3.1	90	2.11
Obscene	651	2.11	163	1.43	26	2.01	61	1.45	52	1.22
Insult	477	1.54	123	1.08	18	1.39	41	0.98	32	0.75
Any	1533	4.96	363	3.19	43	3.32	136	3.24	96	2.25

Table 22: Predictions of the binary classifier model - coronavirus dataset - random forest 100 trees 500 depth

4.5 Discussion and future work

4.5.1 General Analysis

With the Brexit topic, the aim was to show how this detection tool can be used on sensitive topics in society to assess the extent to which a topic, an opinion or a person generates hate. While the polished language of the MPs did not allow us to conclude on this, it did allow us to reflect on the language of a public figure online.

The other two use cases provided insights into the debate discussed in sections 2.1.4 and 2.1.5: which view of free speech is favourable to minimize hate speech online, and whether its legislation can really be effective. The results of our model on tweets about covid and politics led us to the beginnings of a correlation in favour of freedom of expression as legislated in the European Union, i.e. with restrictions. As seen in section 2.1.4, putting some restrictions on freedom of expression can paradoxically strengthen this freedom, or at least limit hate speech propagation.

However, this conclusion must be nuanced. To affirm the existence of a strong correlation, it would be necessary to ensure its existence without language bias or other possible explanations, notably cultural ones.

Moreover, correlation is not causation, and we have too little evidence at this point to say that recent online freedom legislation is resulting in less hateful content online.

4.5.2 Future work

The work done in this essay could be taken further by bringing more extensive insights into the debate on freedom of expression. Using more data, on more topics, and models running in several languages, the existence of this suggested correlation could be further investigated. By considering a temporal evolution, one could also assess the impact of the European Union's measures or of Elon Musk's takeover of Twitter. Moreover, it would allow to reinforce or hinder certain philosophical positions on freedom. Finally, further research would allow to make a case for the existence of causality between restrictions on freedom of expression and the rate of online hate.

It would be interesting to extend this research and this question to other geographical areas, with different cultures and political regimes. Also, it would be interesting to compare the EU and the rest of Europe to study the impact of the EU.

The online hate speech detection tool developed in this project could also be used for analyses on other topics than freedom of speech. For example, it could be used to assess the effectiveness of moderation across multiple platforms, or the link between online hate and hate crime could be used to try to predict future crimes.

Finally, the surprisingly high results of our random dataset might indicate an unseen data bias; in the future, creating ourselves the datasets using Twitter's API would allow us to have better control over eventual biases.

5 Conclusion

The emergence of hate speech on the Internet has given it a new, growing, contagious and dangerous extent. The grave consequences of hate speech are varied; they can be social with the destruction of reputations, psychological with damage to the mental health of victims, and physical with a strong link between online hate and hate crime.

The fight against online hate and its consequences is complex. First, there is a political challenge; different philosophies about freedom of expression clash and create conflicts in legislation and difficulties in applying certain laws within international platforms. Then there is a technical challenge of implementing automatic detection of online hate speech by algorithms.

The creation of these algorithms involves the use of natural language processing and machine learning techniques. Automatic online hate detection has made a lot of progress, but still has room for improvement in language understanding in the future.

I created an online detection model using a public dataset of Wikipedia comments to create six binary classifiers, each capable of detecting one type of hate. Unfortunately, some types of hate had not enough data to provide sufficient performance. Nevertheless, a useful model was trained, which could be improved in the future by implementing more complex techniques like deep learning.

Finally, this detection model was applied to three use cases and a random dataset of tweets. An analysis was made about hate speech on tweets about Brexit, politics, and the coronavirus. These use cases gave insights into the debate on freedom of expression and its legislation, with an emerging correlation in favour of the European Union's vision to reduce hate speech while preserving freedom of speech. However, this conclusion needs to be further developed in the future to be proven and to explore an eventual causal relationship.

6 Appendix

6.1 Databases

Sample screenshots are provided when the raw dataset doesn't have too many columns.

Wikipedia comments Link to dataset: www.kaggle.comcjigsaw-toxic-comment-classification-challengedata

train

id	comment_text	toxic	severe_toxic	obscene	threat	insult	identity_hate
0000997932d777bf	Explanation Why the edits made under my username Hardcore Metallica Fan were reverted? They weren't vandalisms, just closure on some GAs after I voted at New York Dolls FAC. And please don't remove the template from the talk pa	0	0	0	0	0	0
000103f0d9cfb60f	D'aww! He matches this background colour I'm seemingly stuck with. Thanks. (talk) 21:51, January 11, 2016 (UTC)	0	0	0	0	0	0
000113f07ec002fd	Hey man, I'm really not trying to edit war. It's just that this guy is constantly removing relevant information and talking to me through edits instead of my talk page. He seems to care more about the formatting than the actual i	0	0	0	0	0	0
0001b41b1c6bb37e	" More I can't make any real suggestions on improvement - I wondered if the section statistics should be later on, or a subsection of ""types of accidents"" -I think the references may need tidying so that they are all in the exact sar There appears to be a backlog on articles for review so I guess there may be a delay until a reviewer turns up. It's listed in the relevant form eg Wikipedia:Good_article_nominations#Transport "	0	0	0	0	0	0
0001d958c54c6e35	You, sir, are my hero. Any chance you remember what page that's on?	0	0	0	0	0	0
00025465d4725e87	" Congratulations from me as well, use the tools well. · talk "	0	0	0	0	0	0
0002bcb3da6cb337	COCKSUCKER BEFORE YOU PISS AROUND ON MY WORK	1	1	1	0	1	0

Figure 4: Screenshot of raw train dataset

test

id	comment_text
00001cee341fdb12	Yo bitch Ja Rule is more succesful then you'll ever be whats up with you and hating you sad mofuckas...i should bitch slap ur pethedic white faces and get you to kiss my ass you guys sicken me. Ja rule is about pride in da r
0000247867823ef7	== From RfC == The title is fine as it is, IMO.
00013b17ad220c46	" == Sources == * Zawe Ashton on Lapland — / "
00017563c3f7919a	:If you have a look back at the source, the information I updated was the correct form. I can only guess the source hadn't updated. I shall update the information once again but thank you for your message.
00017695ad8997eb	I don't anonymously edit articles at all.
0001ea8717f6de06	Thank you for understanding. I think very highly of you and would not revert without discussion.
00024115d4cbde0f	Please do not add nonsense to Wikipedia. Such edits are considered vandalism and quickly undone. If you would like to experiment, please use the sandbox instead. Thank you. -

Figure 5: Screenshot of raw test dataset

test_labels

id	toxic	severe_toxic	obscene	threat	insult	identity_hate
00001cee341fdb12	-1	-1	-1	-1	-1	-1
0000247867823ef7	-1	-1	-1	-1	-1	-1
00013b17ad220c46	-1	-1	-1	-1	-1	-1
00017563c3f7919a	-1	-1	-1	-1	-1	-1
00017695ad8997eb	-1	-1	-1	-1	-1	-1
0001ea8717f6de06	0	0	0	0	0	0
00024115d4cbde0f	-1	-1	-1	-1	-1	-1
000247e83dcc1211	0	0	0	0	0	0

Figure 6: Screenshot of raw test labels dataset

Random Dataset Link to dataset: www.kaggle.com/datasets/crowdflower/twitter-user-gender-classification

Brexit Dataset Link to dataset: www.kaggle.com/datasets/chadjinik/labelledbrexittweets?select=corpus.csv

corpus

name	screen_name	id	created_at	text	label
Adam Afriyie	AdamAfriyie	741408739983560705	2016-06-10 23:16:32	Enjoyable evening/Q&A wt Chesham&Amersham Conservatives supper club:Our Party in Gov is more important than the EU Ref @BCACF @Conservatives	Leave
Adam Afriyie	AdamAfriyie	739831840064864258	2016-06-06 14:50:30	Science will be enriched & safer outside the EU #Brexit @vote_leave https://t.co/uogxarzVwl	Leave
Adam Afriyie	AdamAfriyie	734678699757436928	2016-05-23 09:33:46	Putting the case for Brexit on CNN #voteleave https://t.co/HSy6auc16o @CNN @camanpour @consforbritain @vote_leave	Leave
Adam Afriyie	AdamAfriyie	732916569173577729	2016-05-18 12:51:41	Science has much to fear from remaining in the EU, but not much to fear from Brexit https://t.co/KMetlR2Vt8	Leave
Adam Afriyie	AdamAfriyie	731237748032557056	2016-05-13 21:40:39	Addressed Datchet VoteLeave EU tonight. Strong turnout. Great Q&A. Young Joseph-year 12-was star of the show. Put up yr posters! @vote_leave	Leave

Figure 7: Screenshot of raw Brexit dataset

#Politics Dataset Link to dataset: www.kaggle.com/datasets/kaushiksuresh147/political-tweets

Coronavirus Dataset This week dataset was made of 7 daily datasets Link to dataset: www.kaggle.com/datasets/smid80/co
covid19-tweets-late-april?select=2020-04-16+Coronavirus+Tweets.CSV

6.2 Access Code

NB: Please be careful using the links. The links are shared links towards google colab and google drive and might reveal my google account, impairing the anonymity of the dissertation.

The code is entirely available and commented here: https://colab.research.google.com/drive/1dO0-yHKy9x4Qu8ZmJlM5QN
47VF?usp=sharing

You can download the code as .ipynb and .py files with the following link: https://drive.google.com/drive/folders/1XiC-PKc5nSYbXm-ZmVczLlOv07LVuNIh?usp=sharing

NLTK library documentation : https://www.nltk.org/

SciKit-Learn library documentation : https://scikit-learn.org/stable/index.html

6.3 Discussions

The discussion with Laurent Rignault was by phone on Friday 8th April 2022 at 10 am. The discussion lasted about one hour. Laurent Rignault gave his verbal consent to be cited in this dissertation and for me to use the information he gave me.

The discussion with Joël Rubinfeld was by phone on Tuesday 8th April 2022 at 6 pm. The discussion lasted about half an hour. Joël Rubinfeld gave his verbal consent to be cited in this dissertation and for me to use the information he gave me.

6.4 Word Counts

Section	N words
Preface	490
Abstract	269
Main body	9988

Table 23: Word counts

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